

ACYANOTIC CONGENITAL HEART DEFECTS

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Classification :

Acyanotic Congestive Heart Defects (CHD)

Left to Right (L → R) shunts

Obstructive lesions :

(Oxygenated $\xrightarrow{\text{mixes}}$ Deoxygenated blood → No cyanosis)

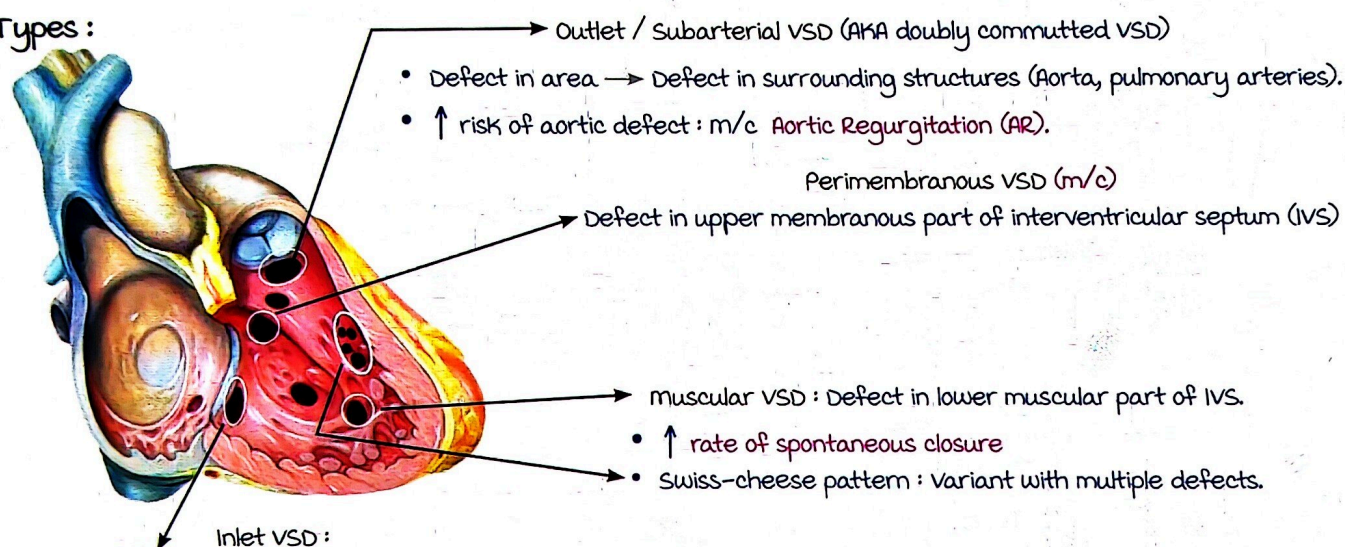
- Coarctation of aorta.

1. Atrial Septal Defect (ASD) : 5-10%.
2. Ventricular Septal Defect (VSD) m/c : 25%.
3. Patent Ductus Arteriosus (PDA).

Ventricular Septal Defects

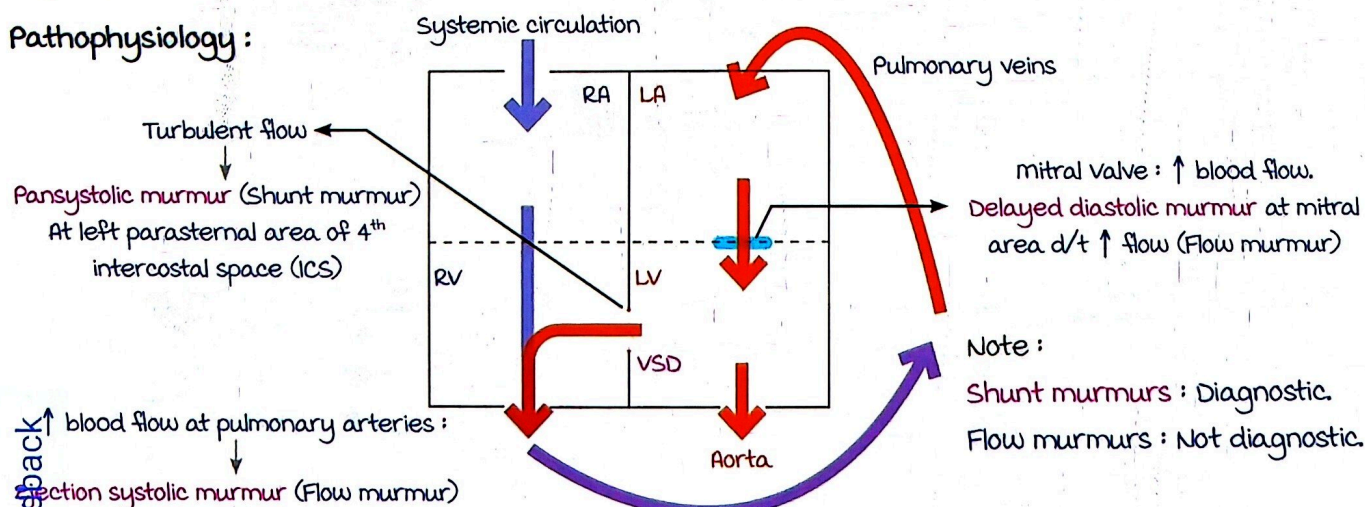
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Types :



- Never close spontaneously.
- ↑ risk of valve defects : m/c mitral valve.

Pathophysiology :



Paediatrics • v1.0 • Marrow 8.0 • 2024

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X-ray findings :

Cardio-thoracic (CT) ratio :

$$CT \text{ ratio} = \frac{a+b}{c}$$

Where :

a+b is sum of widest length on each side.

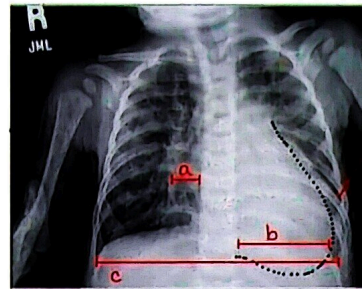
c is widest length of thorax

Normal value → Children < 0.55

→ Infant : < 0.6

- Higher value : Cardiomegaly.

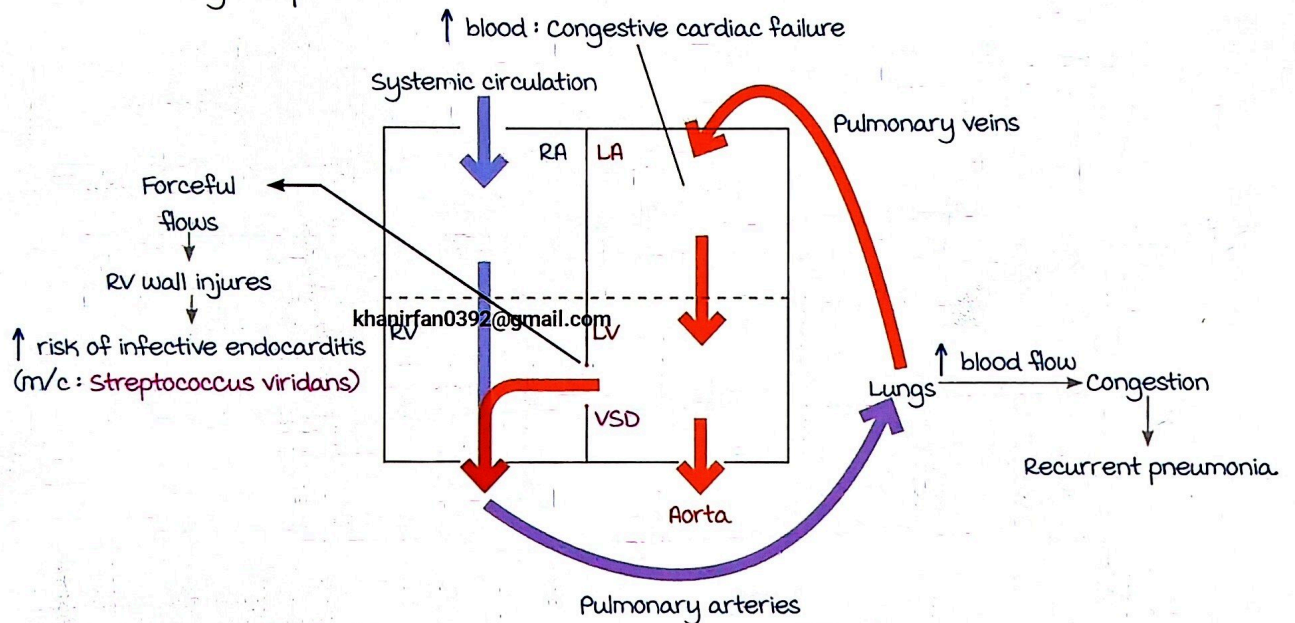
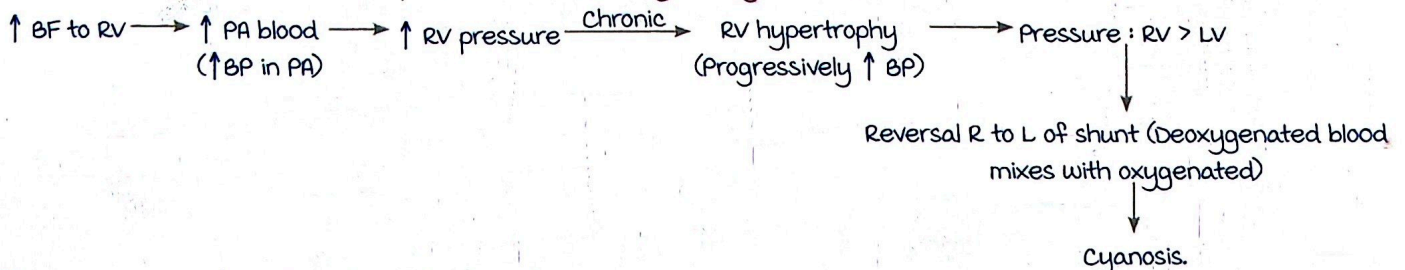
- ↑ CT ratio seen in VSD.



↑ amount of blood to LV
↓
Enlargement of LV
↓
Cardiac outline ↑, apex pushed outwards

Complications :

Early Complications :

**Late complication : Eisenmenger's syndrome****Treatment :**

Medical management (Non-curative) :

- For CCF : Digoxin, Furosemide.
- Supportive (Eg : Antibiotics).

Surgical management (Definitive):

• Indications.

1. CCF (After supportive treatment).
2. $Q_p : Q_s$ (Pulmonary BF : Systemic BF) $> 2:1$.
3. Associated valve defect (Especially AR).

• Surgical options:

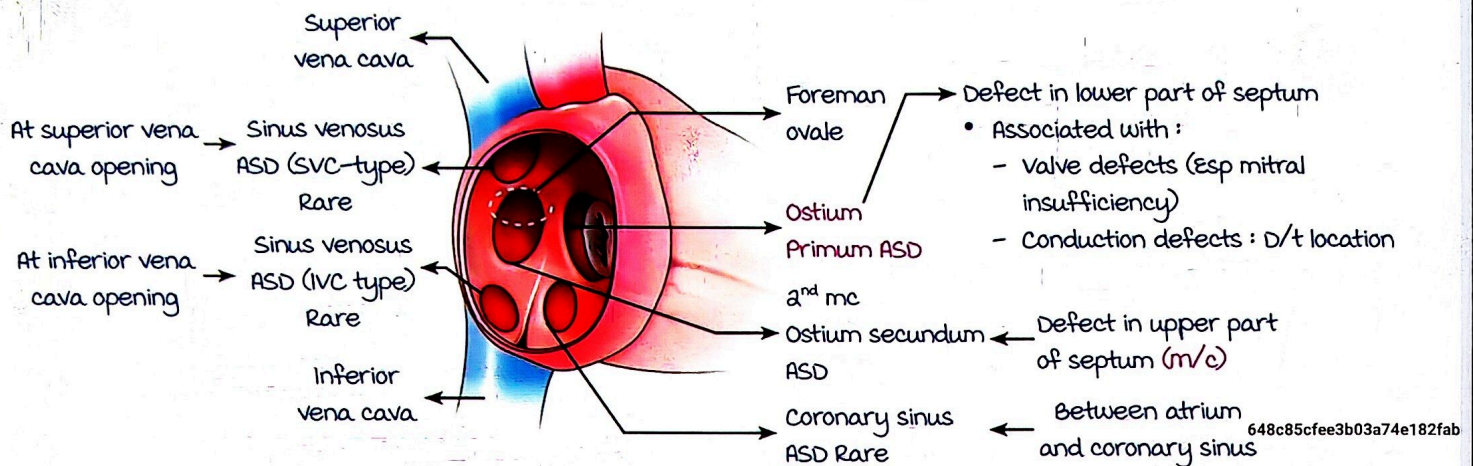
- Closure with dacron patch closure.
- Percutaneous device closure.

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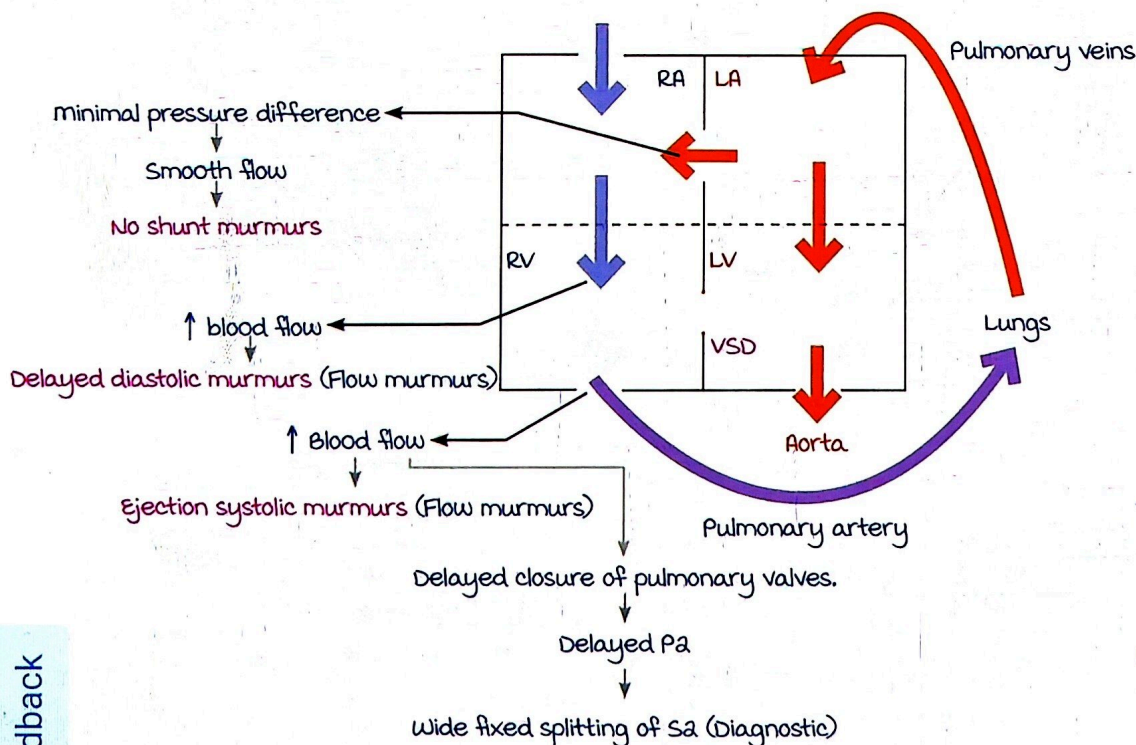
Atrial Septal Defect

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Types:



Pathophysiology:



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ECG findings :

ECG : Done to determine location of ASD.

Ostium secundum ASD : Ostium primum ASD :

Right-axis deviation.

Left-axis deviation.

D/t :

1. Left fascicular block.
2. mitral valve insufficiency → Left chamber overload

TREATMENT :

1. Elective surgical closure :

• Indication :

- ASD > 3-5y (upto school going age)
(No spontaneous closure).
- Size > 8mm.

2. Early surgical closure :

• Indication :

- Uncontrolled congestive cardiac failure.
- Qp : Qs ratio > 2:1.

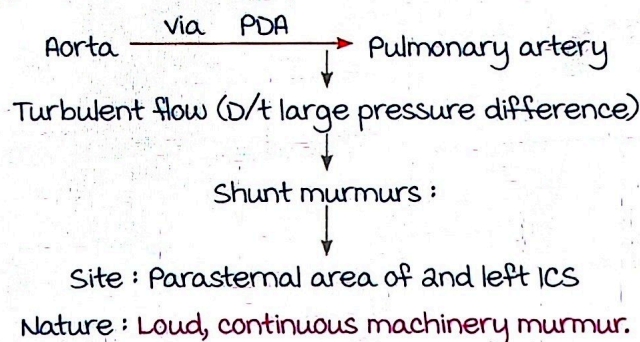
Patent Ductus Arteriosus

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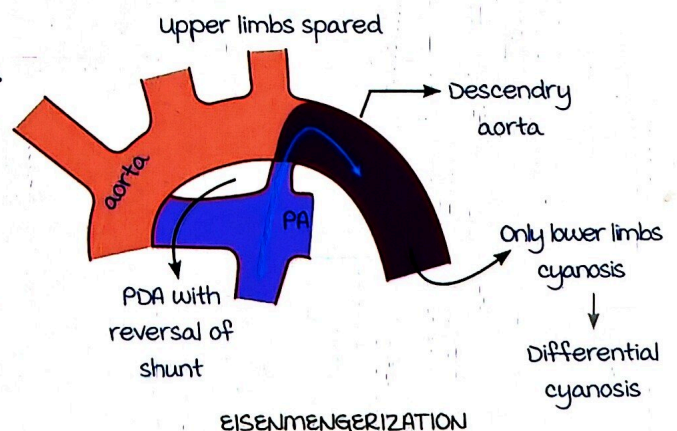
Incidence : Preterm (Kept patent by prostaglandin) > Term babies.

Associations : **Congenital Rubella.****Pathophysiology :**

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**Complications :**

1. Congestion : D/t ↑ blood flow.
 - In lungs (First) → Recurrent pneumonia
 - In heart → CCF.
2. Eisenmenger syndrome :
 - Reversal of shunt (Late complication).



management:**medical:**

- Supportive :
 - CCF : Digoxin, furosemide.
 - Pneumonia : Antibiotics.
- Curative : Only in **preterm** babies.
 - NSAIDs → ↓ prostaglandin production
 - ↓
 - Closure of PDA
 - Ibuprofen > Indomethacin.
 - D/t ↓ risk of
 - Nephrotoxicity
 - Necrotizing enterocolitis (NEC)

Surgical:

• Indications :

1. Preterm baby
 - Not responding to medical mx OR
 - C/I to medical mx
 - Platelet count < 80,000.
 - Serum creatinine > 1.8 mg/dl.
 - Any active bleeding.

2. All term babies.

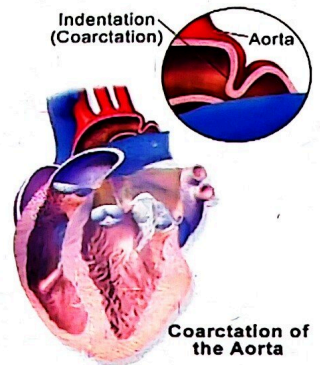
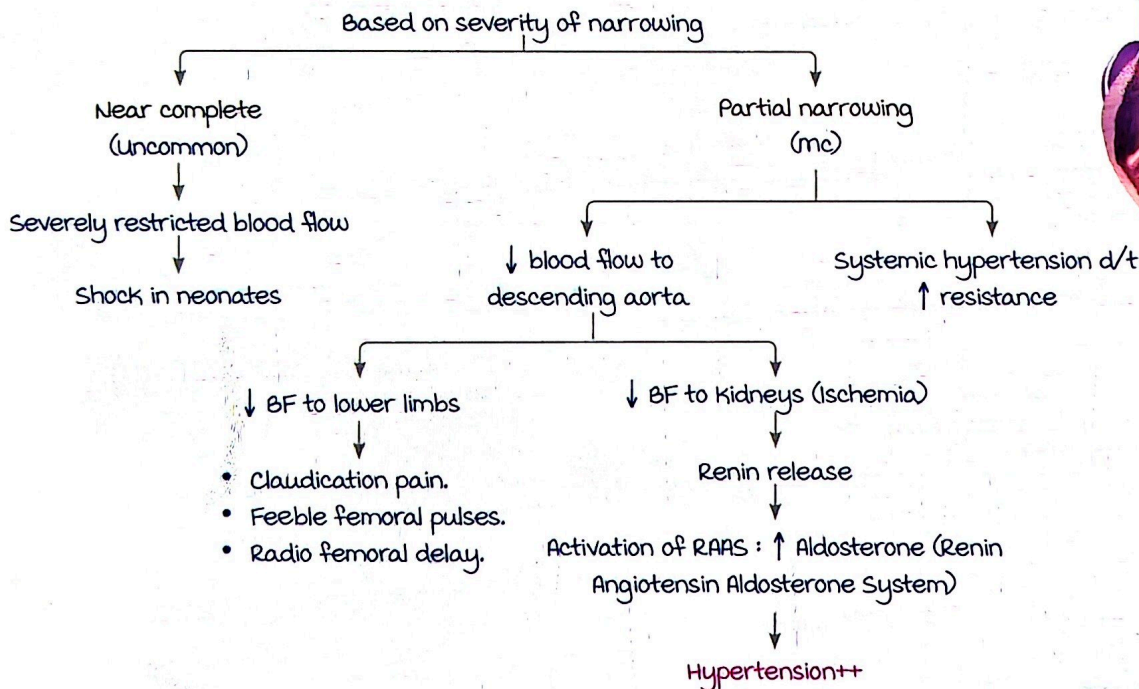
• Technique :

1. Ligation.
2. Percutaneous catheterization using
 - Occluders or
 - Coils

Coarctation of aorta

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m/c location : Juxtaductal (Near ductus arteriosus).

Features :

Note : Hypertension → Hypertensive encephalopathy → Presents with seizures.

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Cardiovascular System

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Compensatory Changes :

Noted in older children (>10y)

Formation of collaterals

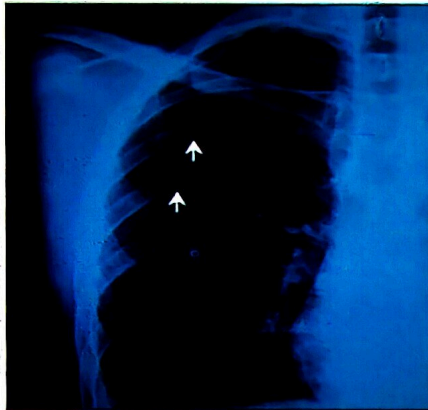
Blood diverted to

Intercostal (IC) arteries

↑ size of IC artery (dilated and tortuous)

Pressure on lower border of rib

Inferior rib notching



Treatment :

Balloon dilatation of coarctation ± Stenting.

